

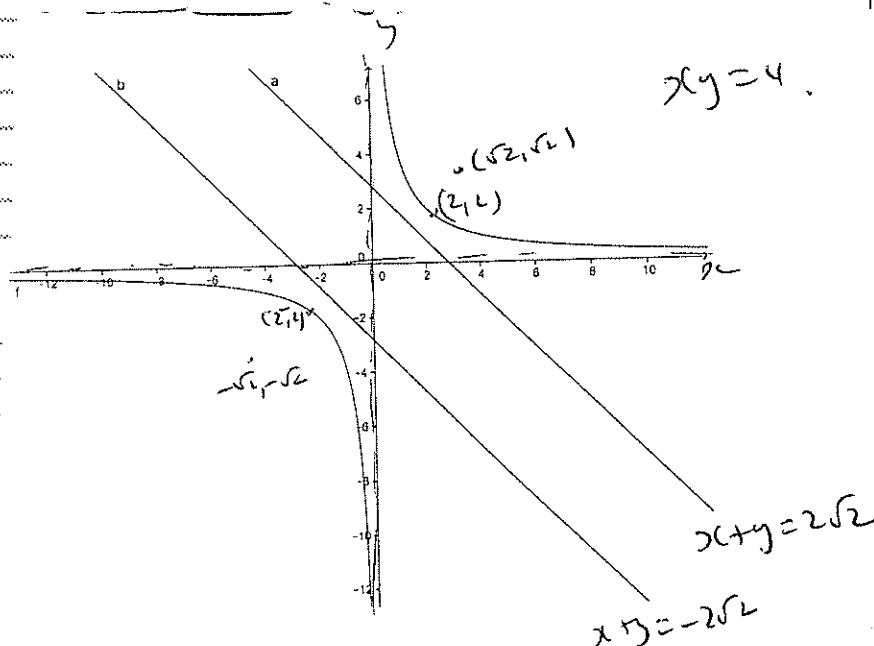
MATHEMATICS Extension 2: Question ... 1...

Suggested Solutions

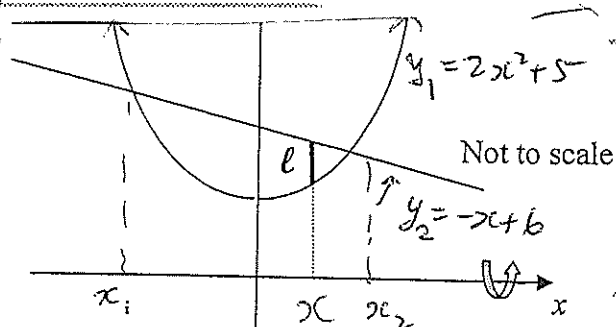
Marks

Marker's Comments

a)



b)



$$(i) \text{ Area} = \pi (y_2^2 - y_1^2) \\ = \pi ((6-x)^2 - (2x^2+5)^2)$$

$$(ii) 6-x = 2x^2+5 \\ 2x^2+x-1=0 \\ (2x-1)(x+1)=0$$

$$x = \frac{1}{2} \text{ or } -1$$

1 for vertex
1 for directrices and asymptotes
1 for foci
1 for graph well done.

Students not drawing slice
more likely to get this incorrect

poor algebra

MATHEMATICS Extension 1 : Question

Suggested Solutions

Marks

Marker's Comments

$$\Delta V = \pi (11 - 12x - 19x^2 - 4x^4) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-1}^{1/2} \pi (11 - 12x - 19x^2 - 4x^4) \Delta x$$

$$V = \int_{-1}^{1/2} \pi (11 - 12x - 19x^2 - 4x^4) dx$$

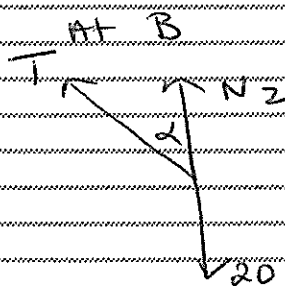
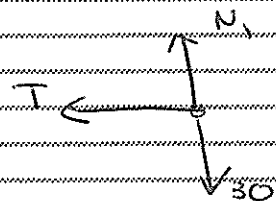
$$= \pi \left[\frac{11x}{1} - \frac{12x^2}{2} - \frac{19x^3}{3} - \frac{4x^5}{5} \right]_{-1}^{1/2}$$

$$= \pi \left[\frac{11}{60} + \frac{148}{15} \right]$$

$$= \frac{261}{20} \pi$$

c) Resolving forces

(i) At A



(ii) Resolving forces at A

$$T = M\omega^2 r \quad \therefore T = 3 \times 2^2 \times r = 12r$$

Resolving forces at B

horizontally: $T \sin \alpha = M\omega^2 R$

vertically: $T \cos \alpha + N_2 = Mg$

Students obtain -1 and $1/2$ but then only went from 0.
poor expansion

1 for each
Some students missed N_1 on A.

MATHEMATICS Extension 1 : Question.....

Suggested Solutions

Marks

Marker's Comments

by equating forces

$$T \sin \alpha = 2 \times (4-r) \sin \alpha \times 4$$

$$T \sin \alpha = 2 \times (4-r) \sin \alpha \times 4$$

$$\therefore T = 8(4-r)$$

$$\text{but } T = 12r$$

$$\therefore 12r = 32 - 8r$$

$$20r = 32$$

$$r = \frac{8}{5}$$

$$\text{So } T = 12 \times \frac{8}{5}$$

$$= 19.2 \text{ N}$$

(ii)

$$T \cos \alpha + N = mg$$

$$N = 20 - T \cos \alpha$$

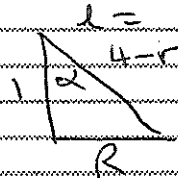
$$= 20 - 19.2 \cos 2$$

$$= 20 - 19.2 \times \frac{1}{12}$$

$$= 12 \text{ N}$$

$$\text{since } l = 4 - 1.6$$

$$= 2.4$$



1

poorly done
those who
tried to
use pythagoras
rather than
trig were
less successful.

1

poorly done.
Students could
have used
part above.

1

1

2a) $y' = \frac{-16}{x^2}$

At $x=2$ $y' = -4$

$m^\perp = \frac{1}{4}$

E& Normal at $(2, 8)$

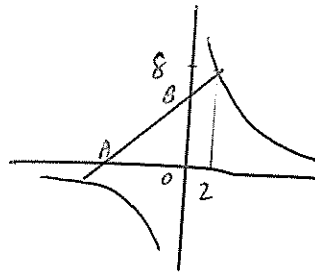
$y - 8 = \frac{1}{4}(x - 2)$

$y = \frac{x}{4} + 1\frac{1}{2}$ or $x - 4y + 30 = 0$

At A $y=0$, $x=-30$

At B $x=0$, $y=1\frac{1}{2}$

Area $\triangle AOB = \frac{1}{2} \times 30 \times 1\frac{1}{2} = 112.5 \text{ unit}^2$



1m

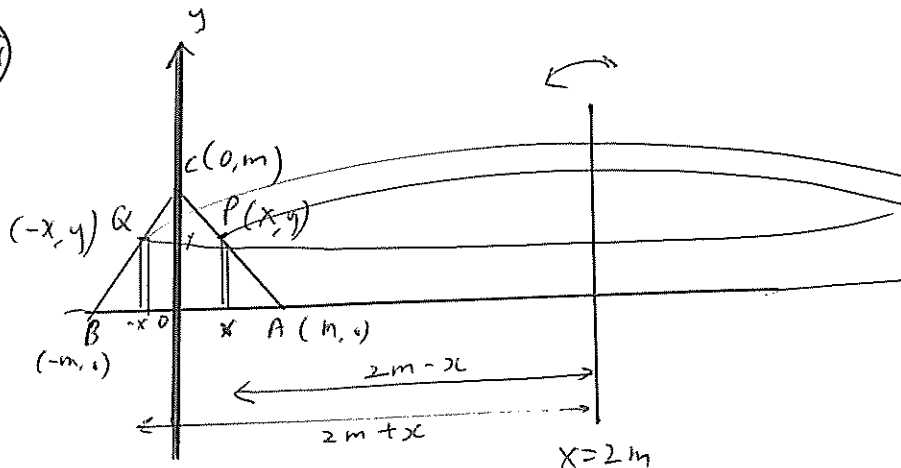
well done.

1m

1m

1m

b)



$\Delta V_P = 2\pi(2m-x)y \Delta x$

$\Delta V_Q = 2\pi(2m+x)y \Delta x$

Assume $x > 0$

1m for diagram

$\Delta V = [2\pi(2m-x)y + 2\pi(2m+x)y] \Delta x$

$= 2\pi \cdot 4my \Delta x$

$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^m 4\pi m 2y \Delta x$

$V = 4\pi m 2 \int_0^m y dx$

$A = \text{Area } \triangle ABC = 2 \int_0^m y dx$ ($\triangle ABC$ is symmetrical about y-axis)

$V = 4\pi m A$

ii) Area $\triangle ABC = \frac{1}{2} \cdot 2m \cdot m = m^2$

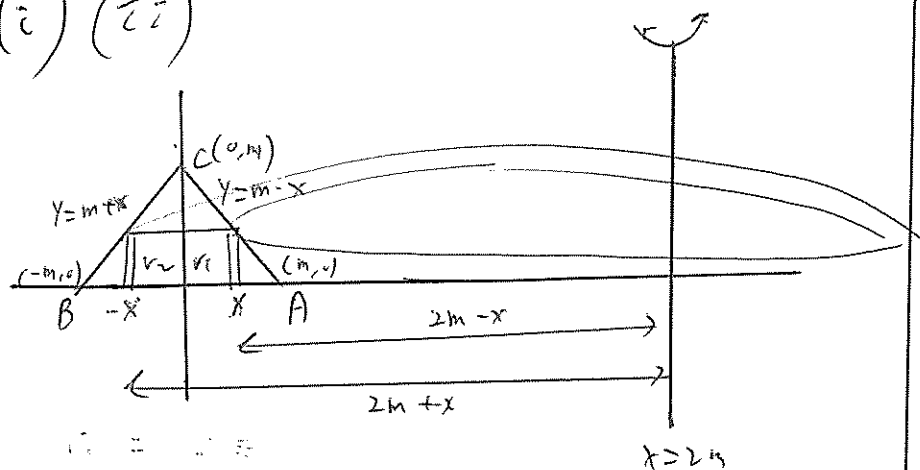
$\therefore \text{Vol of solid formed} = 4\pi m m^2 = 4\pi m^3 \text{ unit}^3$

1m

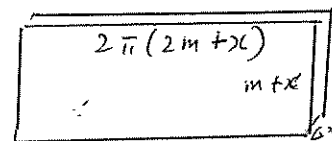
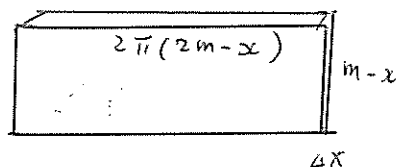
b (i) (2i)

Ex 2

P2



Assume $x > 0$



$$\Delta V_1 = 2\pi(2m-x)(m-x)$$

$$\Delta V_2 = 2\pi(2m+x)(m+x)$$

$$V_1 = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^m 2\pi(2m-x)(m-x) \Delta x$$

$$V_1 = 2\pi \int_0^m (2m^2 - 3mx + x^2) dx$$

$$V_1 = 2\pi \left(2m^2x - \frac{3m}{2}x^2 + \frac{x^3}{3} \right) \Big|_0^m$$

$$V_1 = 2\pi \left(2m^3 - \frac{3}{2}m^3 + \frac{m^3}{3} \right) = \frac{5\pi}{3}m^3$$

Similarly

$$V_2 = 2\pi \int_{-m}^0 (2m-x)(m+x) dx$$

$$= 2\pi \int_{-m}^0 (2m^2 + mx - x^2) dx$$

$$= 2\pi \left(2m^2x + \frac{m}{2}x^2 - \frac{x^3}{3} \right) \Big|_{-m}^0$$

$$= \frac{7\pi}{3}m^3$$

$$\therefore V_1 + V_2 = \frac{5\pi}{3} + \frac{7\pi}{3} m^3 = 4\pi m^3$$

$$\therefore \text{Vol. of the solid formed} = 4\pi m^3$$

$$A = \text{Area of } \triangle ABC = \frac{1}{2}(2m)m = m^2$$

$$\therefore \text{vol of solid formed} = \underline{\underline{4\pi m^3}}$$

A lot of students have tried to use this method but failed to see the limits of V_1, V_2 are different.
 $\therefore$ failed to arrive at the correct answer.

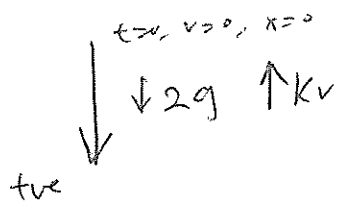
(1m) for $V_1 + V_2 = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^m 2\pi(2m-x)(m-x) \Delta x + \lim_{\Delta x \rightarrow 0} \sum_{x=-m}^0 2\pi(2m-x)(m+x) \Delta x$

or 1m for $\frac{5\pi}{3}m^3$ or $\frac{7\pi}{3}m^3$ together with $|ABC| = m^2$

(1m) for each $4\pi m^3$

(1m) for $4\pi m^3$

c i)



1 m for diagram
with the
direction

ii) $m\ddot{x} = mg - kv$ (Newton's Law of Motion)
 $2\ddot{x} = 2g - kv$ ($m=2$)

must show
 $m=2$

$$\ddot{x} = g - \frac{kv}{2}$$

Terminal velocity when $\ddot{x} = 0$

$$0 = g - \frac{kv}{2}$$

$$v_T = \frac{2g}{k} \text{ n/s} \quad \#$$

1 m

iii) $\frac{dv}{dt} = \frac{2g - kv}{2}$

1 m

$$\int_0^v \frac{dv}{2g - kv} = \int_0^t \frac{dt}{2}$$

$$-\frac{1}{k} \ln(2g - kv) \Big|_0^v = \frac{t}{2}$$

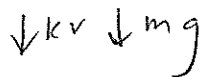
$$-\frac{1}{k} \ln\left(\frac{2g - kv}{2g}\right) = \frac{t}{2}$$

$$t_d = \frac{-2}{k} \ln\left(\frac{2g - kv}{2g}\right) \quad \therefore e^{-\frac{kt}{2}} = 1 - \frac{kv}{2g}$$

$$v = \frac{2g}{k} \left(1 - e^{-\frac{kt}{2}}\right) \quad \#$$

1 m

(many forget v
expression)

2nd particle

$$2\ddot{x} = -(2g + kv)$$

Newton's Law

$$\ddot{x} = -\left(\frac{2g + kv}{2}\right)$$

$$\int_u^0 \frac{2 dv}{-(2g + kv)} = \int_0^t dt$$

(At rest $v=0$)

1 m

$$t = \left. -\frac{2}{k} \ln(kv + 2g) \right|_u^0 \quad \text{Exp 2}$$

$$t_u = -\frac{2}{k} \ln\left(\frac{2g}{ku + 2g}\right)$$

$$t_d = t_u$$

$$-\frac{2}{k} \ln\left(\frac{2g - kv}{2g}\right) = -\frac{2}{k} \ln\left(\frac{2g}{ku + 2g}\right)$$

$$\frac{2g - kv}{2g} = \frac{2g}{ku + 2g}$$

$$4g^2 = 2kug - k^2uv + 4g^2 - 2kv^2$$

$$uvk^2 + 2kv^2 = 2kug$$

$$uv + 2\frac{v^2}{k} = \frac{2ug}{k}$$

$$\text{Let } V_T = \frac{2g}{k}$$

$$\therefore uv + V_T v = V_T u$$

$$\therefore v(u + V_T) = uV_T$$

$$v = \frac{uV_T}{u + V_T}$$

$$\text{Let } V = V_T$$

$$\therefore v = \frac{uV}{u + V}$$

Im

$$t_d = t_u$$

Im

Note Some students

$$\text{wrote } t_d = -\frac{1}{k} \ln\left(\frac{2g - kv}{2g}\right)$$

$$t_u = -\frac{1}{k} \ln\left(\frac{2g}{ku + 2g}\right)$$

Hence double mistake will correct itself in prove

$$v = \frac{uV}{u + V}$$

max get $\frac{3}{5}$

MATHEMATICS Extension 2: Question... 3..

Suggested Solutions

Marks

Marker's Comments

$$(i) \quad y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

when $x = cp$, $m_T = \frac{-c^2}{c^2 p^2} = -\frac{1}{p^2}$

eqn. of tangent: $y - y_1 = m(x - x_1)$

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$y = -\frac{1}{p^2}x + \frac{c}{p} + \frac{c}{p}$$

$$p^2 y = -x + 2cp$$

$$x + p^2 y = 2cp \quad \text{--- (1)}$$

Similarly tangent at Q is $x + q^2 y = 2cq \quad \text{--- (2)}$

solve (1) & (2)

$$x\left(\frac{1}{p^2} - \frac{1}{q^2}\right) = 2c\left(\frac{1}{p} - \frac{1}{q}\right)$$

$$x\left(\frac{1}{p} + \frac{1}{q}\right) = 2c$$

$$x\left(\frac{q+p}{pq}\right) = 2c$$

$$\therefore x = \frac{2cpq}{p+q} \quad \text{--- (1)}$$

$$\therefore y = -\frac{1}{p^2} \left(\frac{2cpq}{p+q} \right) + \frac{2c}{p}$$

$$= \frac{-2cq + 2cp + 2cq}{p(p+q)}$$

$$= \frac{2c}{p+q} \quad \text{--- (1)}$$

(ii) eqn. of chord PQ is

$$y - y_1 = m(x - x_1)$$

$$y - \frac{c}{p} = \frac{\frac{c}{q} - \frac{c}{p}}{cp - cq}(x - cp)$$

$$y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$$

$$pqy - cq = -x + cp$$

$$pqy + x = cp + cq \quad \text{--- (1)}$$

$$pqy + x = c(p+q)$$

had to show working, i.e. derive the formula

(1)

(1)

(1)

(1)

had to show working, if you just quoted the formula you got 0mks.

(1)

2/3

MATHEMATICS Extension 2: Question 3 continued.

Suggested Solutions	Marks	Marker's Comments
(iii) Eqn of OT is $y - 0 = \frac{2c}{2pq}(x - 0)$ $y = \frac{x}{pq}$ $pqy = x$ R is intersection of OT and PQ $pqy = -pqy + c(p+q)$ $2pqy = c(p+q)$ $y = \frac{c(p+q)}{2pq}$ $\therefore x = pq \times \frac{c(p+q)}{2pq}$ $x = \frac{c(p+q)}{2}$ $\therefore R \left(\frac{c(p+q)}{2}, \frac{c(p+q)}{2pq} \right)$		* a number of students didn't use $O(0,0)$ as the (x_1, y_1) in $\textcircled{1}$
(iv) Midpoint of PQ is $\left(\frac{cp+cq}{2}, \frac{\frac{c}{p} + \frac{c}{q}}{2} \right)$ $= \left(\frac{c(p+q)}{2}, \frac{cq+cp}{2pq} \right)$ $= \left(\frac{c(p+q)}{2}, \frac{c(p+q)}{2pq} \right)$ $= R$ $\therefore R$ bisects PQ		$\textcircled{1}$
(b)(i) $a_T = r\ddot{\theta}$ $a_N = r(\dot{\theta})^2$	$\textcircled{1}$	
	$\textcircled{1}$	

MATHEMATICS Extension 2: Question 3... continued.

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 (11) \quad \dot{s} &= r\dot{\theta} \\
 &= r \frac{d\theta}{dt} \\
 &= r \frac{d\theta}{d\theta} \times \frac{d\theta}{dt} \longrightarrow \\
 &= r \dot{\theta} \frac{d\theta}{d\theta} \\
 &= r \left(\frac{d(\frac{1}{2}\theta^2)}{d\theta} \right) \cdot \frac{d\theta}{dt} \\
 &= r \frac{d(\frac{1}{2}\theta^2)}{d\theta} \\
 &= \frac{r}{r^2} \frac{d(\frac{1}{2}v^2)}{d\theta} \\
 &= \frac{d(\frac{1}{2}v^2)}{r d\theta}
 \end{aligned}$$

$$\begin{aligned}
 v &= r\dot{\theta} \\
 \frac{v}{r} &= \dot{\theta}
 \end{aligned}$$

necessary line!!!

①

need to see this
(if its missing = 0 mks)
①

* Don't use ω
because it is not a
VARIABLE!!

$$\begin{aligned}
 (11)(2) \quad F_T &= m\dot{s} \\
 &= mr\dot{\theta}
 \end{aligned}$$

$$\begin{aligned}
 -0.1 &= 0.04 \times 3 \times \dot{\theta} \\
 \dot{\theta} &= -\frac{5}{6}
 \end{aligned}$$

①

$$(12) \quad \dot{s} = \frac{d(\frac{1}{2}v^2)}{r d\theta}$$

$$\text{now, } r\dot{\theta} = 3 \times -\frac{5}{6} = -\frac{5}{2}$$

$$r\dot{\theta} = \frac{d(\frac{1}{2}v^2)}{ds}$$

$$\int_5^0 \frac{d(\frac{1}{2}v^2)}{ds} ds = \int_0^{-5} -\frac{5}{2} ds$$

①

$$\left[\frac{1}{2}v^2 \right]_5^0 = -\frac{5}{2} s$$

$$-\frac{25}{2} = -\frac{5}{2} s$$

$$s = 5$$

$$\text{distance} = 5\text{m}$$

①

MATHEMATICS Extension 2: Question 4....

Suggested Solutions

Marks

Marker's Comments

(i) $m \frac{dv}{dt} = -mk(c+v^2)$
 $\frac{dv}{c+v^2} = -k \, dx$

①

Must show " - " sign as it is a resistive force

(ii) $v \frac{dv}{dx} = -k(c+v^2)$

①

Integral - integral should be shown.

$\int_u^0 \frac{v}{c+v^2} = -k \int_0^D dx$

$\frac{1}{2} \left[\ln[c+v^2] \right]_u^0 = -kD$ (i)

$-\frac{1}{2}k \ln \left[\frac{c}{c+u^2} \right] = D$

①

Correct answer with working as answer was given.

$D = \frac{1}{2k} \ln \left[\frac{c+u^2}{c} \right]$ (ii)

(iii) From (i) $-\frac{kD}{2} = \frac{1}{2} \left(\ln[c+v^2] \right)_{u/2}^{u/2}$

OR. $\left[-\frac{kD}{2} = \frac{1}{2} \left[\ln(c+v^2) \right]_{u/2}^0 \right]$

← ALTERNATIVE

$\frac{D}{2} = \frac{-1}{2k} \left[\ln \left(\frac{c+u^2/4}{c} \right) \right]$

$= \frac{1}{2k} \left[\ln \left(\frac{4(c+u^2)}{4c+u^2} \right) \right]$

$= \frac{1}{2k} \left[\ln \left(\frac{4(c+u^2)^2}{4c+u^2} \right) \right]$ (iii)

Equate logs.

$\frac{c+u^2}{c} = \frac{16(c+u^2)^2}{4c+u^2}$

$\frac{1}{c} = \frac{16(c+u^2)}{4c+u^2}$

$16c^2 + 16u^2c = 16c^2 + 8cu^2 + u^4$

$8cu^2 = u^4 \quad u \neq 0$

$8c = u^2$

①

Several methods possible but must relate D and $D/2$ and remove logs

①

Correct answer.

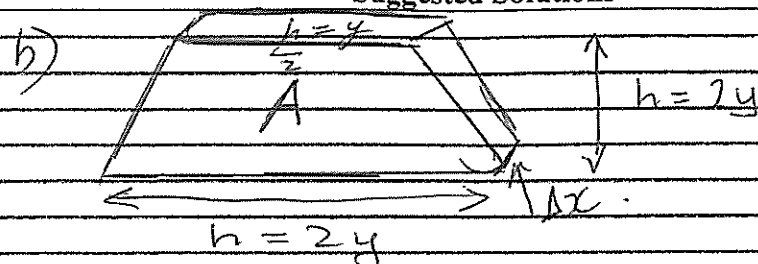
$D = \frac{1}{2(0.01)} \ln \left[\frac{c+8c}{c} \right] = 5 \ln 9$
 $k=0.01 \quad = 10 \ln 3$
 ≈ 10.99

MATHEMATICS Extension 2: Question...4....

Suggested Solutions

Marks

Marker's Comments



$$x^2 + y^2 = 400$$

$$y = \sqrt{400 - x^2}$$

$$\Delta V = \frac{2y}{3} (2y + y) \Delta x$$

$$= 3y^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{-20}^{20} 3y^2 \Delta x$$

$$V = \int_{-20}^{20} 3(400 - x^2) dx$$

Even Function

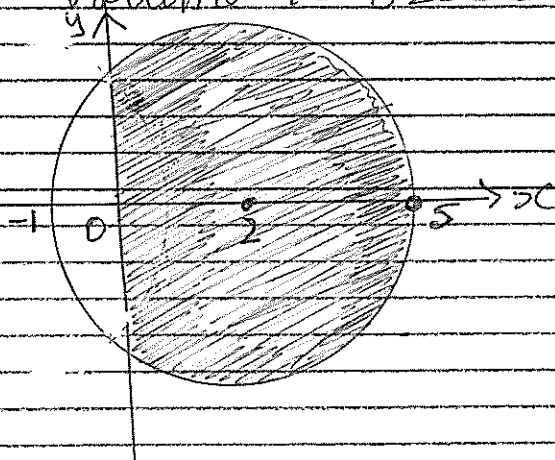
$$V = 2 \times 3 \int_0^{20} (400 - x^2) dx$$

$$= 6 \left[400x - \frac{x^3}{3} \right]_0^{20}$$

$$= 6 \left[8000 - \frac{8000}{3} \right]$$

$$= 32000$$

Volume is 32000 m²



①

$$\Delta V = 3y^2 \Delta x$$

①

$$\text{sub } y = \sqrt{400 - x^2}$$

①

integration

①

answer.

①

correct graph and shaded area.

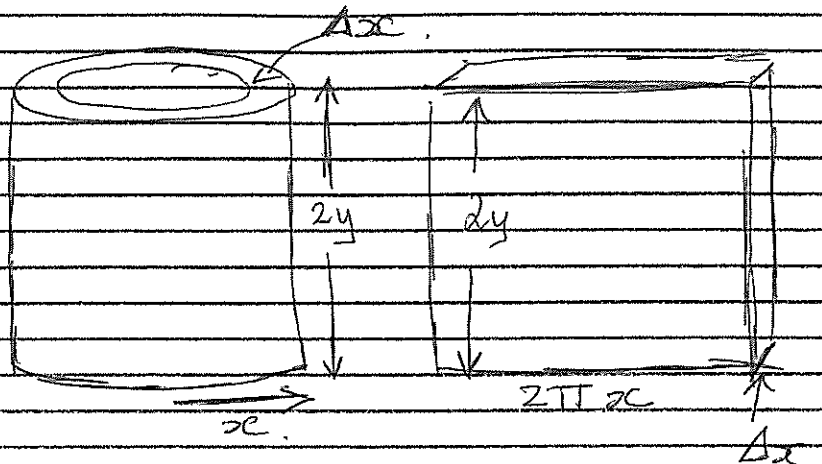
MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

(ii)



$$\Delta V \approx 2\pi x (2y) \Delta x$$

$$\text{OR } \Delta V = 2\pi((x+\Delta x)^2 - x^2)2y$$

$$y^2 = 9 - (x-2)^2$$

$$y = \sqrt{9 - (x-2)^2}$$

$$\Delta V \approx 2\pi x 2\sqrt{9 - (x-2)^2} \Delta x$$

$$= 4\pi x \sqrt{9 - (x-2)^2} \Delta x$$

$$(ii) \quad V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^5 4\pi x \sqrt{9 - (x-2)^2} \Delta x$$

$$V = 4\pi \int_0^5 x \sqrt{9 - (x-2)^2} dx$$

$$\text{Let } (x-2) = 3 \sin \theta \quad 3 \cos \theta = \sqrt{9 - (x-2)^2}$$

$$\frac{dx}{d\theta} = 3 \cos \theta$$

$$x=5 \quad \sin \theta = 1 \quad \theta_1 = \pi/2$$

$$x=0 \quad \sin \theta = -2/3 \quad \theta_2 = \sin^{-1}(-2/3)$$

$$V = 4\pi \int_{\theta_2}^{\theta_1} (3 \sin \theta + 2) \times 3 \cos \theta \times 3 \cos \theta d\theta$$

$$= 4\pi \int_{\theta_2}^{\theta_1} 27 (\sin \theta \cos^2 \theta) + 18 \cos^2 \theta d\theta$$

$$= 4\pi \int_{\theta_2}^{\theta_1} -27 (-\sin \theta \cos^2 \theta) + 18 \left(\frac{1}{2} (1 + \cos 2\theta) \right) d\theta$$

$$= 4\pi \left[\frac{-27}{3} \cos^3 \theta + 9\theta + \frac{9}{2} \sin 2\theta \right]_{\sin^{-1}(-2/3)}^{\pi/2}$$

$$= 4\pi \left[9\pi/2 + \frac{5\sqrt{5}}{3} + 9\sin^{-1}(2/3) + \frac{9}{2} \left(\frac{-2}{3} \right) \left(\frac{\sqrt{5}}{3} \right) \right]$$

$$= 18\pi^2 + 44\frac{\sqrt{5}}{3}\pi + 36\sin^{-1}(2/3)$$

$$\approx 363.21$$

①

Must show a diagram and working

①

Substitute y expression.

①

substitution

①

simplify and limit values

* note $\cos \theta_2 = \sqrt{5}/3$

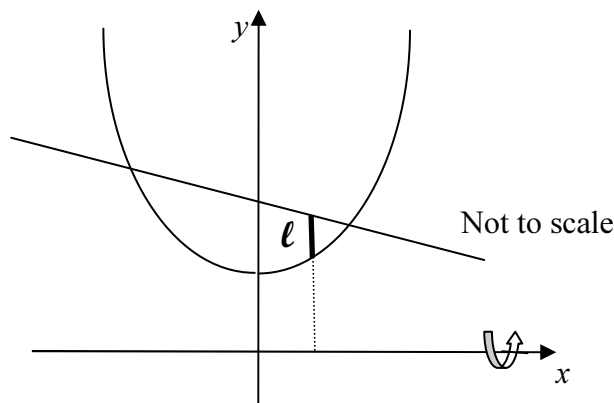
①

correct answer

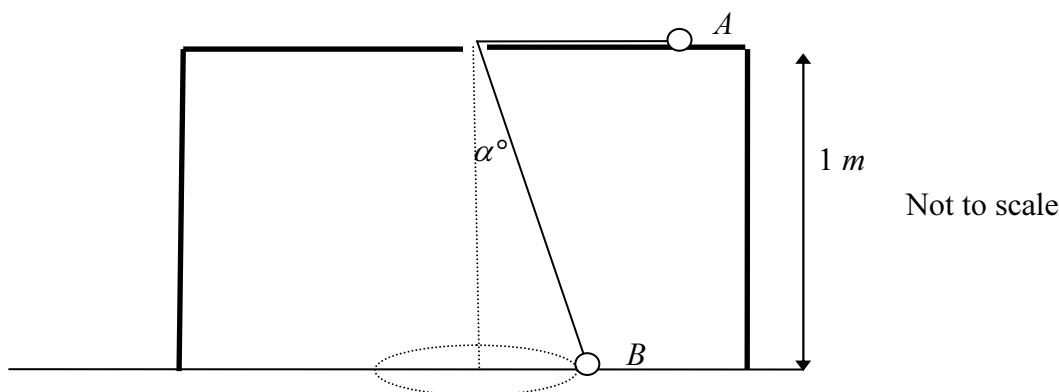
QUESTION 1 (15 Marks)

Marks

- (a) Neatly draw the graph of the curve $xy = 4$.
Show clearly the vertices, foci and the equations of the directrices and asymptotes. 4
- (b) The area bounded by the curves $y = 2x^2 + 5$ and $x + y - 6 = 0$ is rotated about the x -axis as shown in the diagram below. When the region is rotated, the vertical line segment ℓ , at a distance x from the y -axis, sweeps out an annulus.



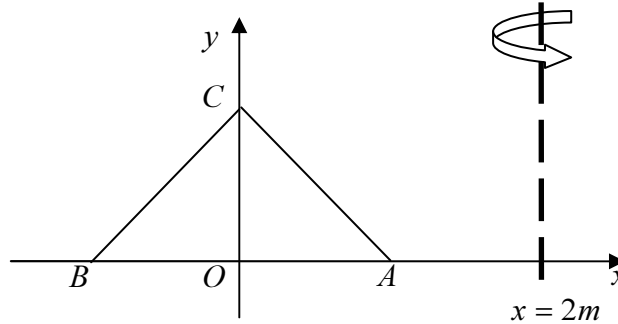
- (i) Find an expression for the area of the annulus swept out by ℓ . 1
- (ii) Hence calculate the volume of the solid of revolution. 3
- (c) Two masses A and B , 3 kg and 2 kg respectively, are attached by a 4 m light inextensible string which passes through a small hole in a smooth horizontal table. Mass A moves in a circle on the table at a uniform angular speed of 2 radians/sec. Mass B moves as a conical pendulum with the same angular speed as mass A and is in contact with the smooth floor so that the string remains taut. (take $g = 10 \text{ m/s}^2$.)



- (i) Draw force diagrams for masses A and B . 2
- (ii) Show that the tension in the string is 19.2 N . 3
- (iii) Calculate the value of the normal reaction that the floor exerts on mass B , if the table is 1 m above the floor. 2

QUESTION 2 (15 Marks) START A NEW PAGE**Marks**

- (a) (i) Find the equation of the normal to the curve $xy = 16$ at the point when $x = 2$. 2
- (ii) This normal cuts the x -axis at A and the y -axis at B . Find the area of the $\triangle AOB$, where O is the origin. 2
- (b) $\triangle ABC$ has vertices $A(m, 0)$, $B(-m, 0)$ and $C(0, m)$ as shown in the diagram below.



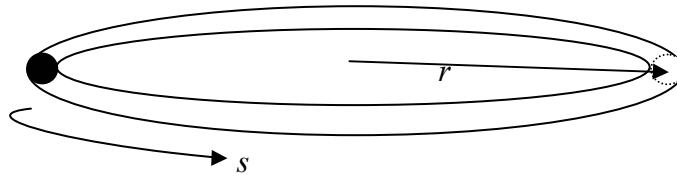
The triangle is rotated about the line $x = 2m$.

- (i) Use cylindrical shells to show that the volume of the solid of rotation is given by $V = 4\pi mA$, where A is the area of $\triangle ABC$. 3
- (ii) Hence find the volume of the solid formed. 1
- (c) A particle of mass 2 kg falls vertically from rest, from a point P , through a substance with resistance of kv Newton, where k is a positive constant and v m/s is the velocity of the mass at time t seconds.
- (i) Draw a diagram showing all forces acting on the particle. 1
- (ii) Show that the terminal velocity of the particle is $V = \frac{2g}{k}$ m/s, 1
where g is the acceleration due to gravity.
- (iii) Find an expression for the velocity of the particle at time t . 2
- (iv) At the same time that the first mass was released, an equal particle is projected vertically upwards from point P with an initial velocity U m/s, in the same substance. 3

Show that the velocity of the first mass when the second mass is momentarily

at rest is given by $v = \frac{VU}{(V+U)}$ where V is the terminal velocity of the first particle.

- (a) (i) Prove that the point of intersection T of the tangents at $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, on the hyperbola $xy = c^2$, is $T\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$. 3
- (ii) Find the equation of the chord PQ . 1
- (iii) The line through points O and T meets the chord PQ at R , where O is the origin. Find the coordinates of R . 3
- (iv) Show that R bisects the interval PQ . 1
- (b) The diagram shows a particle moving in a circular path of radius r metres with variable angular velocity $\dot{\theta}$ rad/sec. .



- (i) Write down the tangential and normal components of the acceleration for the particle in terms of $\dot{\theta}$ and $\ddot{\theta}$. 2
- (ii) When the particle has moved through an arc length of s metres and an angular distance of θ radians, show that 2
- $$\ddot{s} = \frac{d\left(\frac{1}{2}v^2\right)}{r d\theta} \quad \text{where } v \text{ m/s is the speed of the particle.}$$
- (iii) A small ball of mass 40 g slides around in a narrow horizontal circular tube of radius 3 m. The initial speed of the particle is 5 m/s and there is a frictional resistance force in the tube of 0.1 N.
- (α) Write down the equation of motion of the ball in terms of $\ddot{\theta}$. 1
- (β) Calculate the distance travelled by the ball before coming to rest. 2

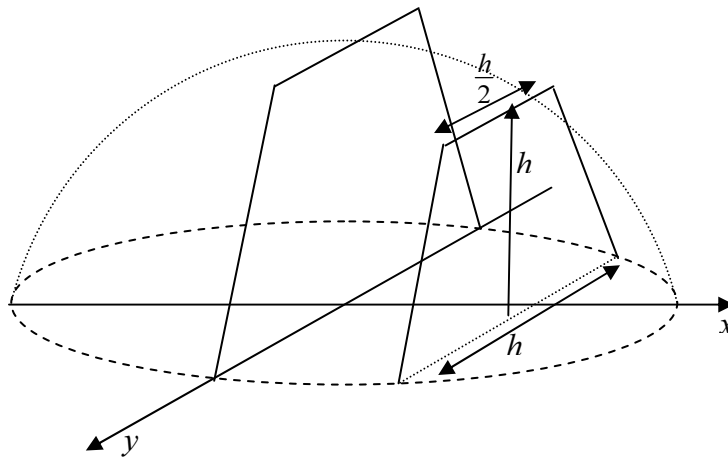
QUESTION 4 (15 Marks)**Marks**

- (a) A resistance force of $mk(c + v^2)$ N acts directly against the motion of a particle moving in a horizontal straight line. The mass of the particle is m kg, v m/s is its velocity and k and c are positive constants. Initially the particle has a velocity U m/s ($U > 0$) and comes to rest after it has travelled D metres.

When it has travelled half this distance its speed is $\frac{U}{2}$ m/s.

- (i) Write the equation of motion for the particle. 1
- (ii) Show that $D = \frac{1}{2k} \log_e \left(\frac{c + U^2}{c} \right)$ 2
- (iii) Find the total distance D covered when $k = 0.1$. 2

- (b) A storage shed is built with a horizontal circular base of radius 20 m. Vertical cross-sections are trapezia with heights equal to their base lengths and tops half the base length as shown in the diagram below.



Calculate the volume to the storage shed. 4

- (c) (i) Shade the region represented by $(x - 2)^2 + y^2 \leq 9$ for $x \geq 0$ on a number plane 1
- (ii) The shaded region described above is rotated about the y -axis, Use the method of cylindrical shells to show that $\Delta V \approx 4\pi x \sqrt{9 - (x - 2)^2} \Delta x$ for a cylinder of radius x units. 2
- (iii) Hence find the volume of the solid formed. 3

END OF EXAMINATION